

Edge AI-Based Smart Traffic Signal Control: A Proposed Architecture for Real-Time Traffic Management

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Abstract—Traffic signal control is an important component of intelligent transportation systems. Conventional traffic signals generally operate using predefined timing plans and may have limited ability to respond to rapidly changing traffic conditions. Recent research on Edge Artificial Intelligence (Edge AI) indicates that processing data closer to its source can support real-time decision-making while reducing dependence on centralized cloud processing. This paper proposes a conceptual Edge AI-based architecture for adaptive traffic signal control. The proposed system uses roadside cameras or sensors to collect traffic information, an edge computing unit to process the data locally, and a traffic signal controller to apply the resulting signal decision. Cloud infrastructure is considered for optional storage, monitoring, and large-scale analysis rather than for every immediate control decision. The paper discusses the system architecture, processing flow, traffic-state representation, implementation considerations, security requirements, and limitations. No experimental performance values are claimed because the proposed architecture has not been experimentally evaluated in this sample study.

Index Terms—Edge AI, Edge Computing, Artificial Intelligence, Intelligent Transportation Systems, Traffic Signal Control, Machine Vision, Smart Traffic Management

I. INTRODUCTION

The growth of connected devices, artificial intelligence (AI), and edge computing has created new opportunities for real-time intelligent systems. Edge AI combines AI with computation performed close to the location where data is generated. This approach is useful for applications that require rapid processing and decision-making [1].

Traditional cloud-based systems generally transmit data to remote computing infrastructure for processing. Although cloud computing provides substantial processing and storage capabilities, transmitting large volumes of sensor or video data can create communication and bandwidth requirements. Edge computing moves processing closer to the data source, which can reduce the amount of information that must be transmitted to a remote server [1].

Transportation is one of the application areas identified for Edge AI. The reviewed literature discusses traffic-light operations and edge-based traffic data processing as potential applications [1], [2]. AI-based approaches have also been

studied for traffic signal control, including machine vision and reinforcement-learning-based approaches [3], [4].

Based on these observations, this paper presents a proposed architecture for Edge AI-based traffic signal control. The goal is not to claim measured improvement, but to describe a technically plausible architecture that can be implemented and evaluated in future work.

II. PROBLEM STATEMENT

Traffic conditions can change continuously because of variations in vehicle density, road demand, incidents, and other factors. A fixed signal timing strategy may not respond directly to every change in traffic conditions.

Cloud-based AI can provide centralized computing resources, but traffic-control applications may require processing close to the road. Therefore, there is a need to investigate an architecture in which traffic information can be processed locally before control decisions are communicated to traffic signals.

The proposed problem can therefore be stated as follows:

How can Edge AI be used to process local traffic information and support adaptive traffic signal control while considering latency, bandwidth, resource limitations, security, and scalability?

III. OBJECTIVES

The main objectives of the proposed system are:

- 1) To collect traffic information using roadside cameras or sensors.
- 2) To process traffic information near the data source using an edge computing device.
- 3) To represent the observed traffic condition using suitable traffic-state variables.
- 4) To generate a signal-control decision based on the estimated traffic condition.
- 5) To reduce unnecessary transmission of raw traffic data to centralized servers.
- 6) To provide a conceptual architecture that can be experimentally evaluated in future work.

IV. RELATED WORK

A. Edge AI

Edge AI refers to the integration of AI capabilities with computing resources located near data sources. The reviewed Edge AI literature describes its importance for real-time processing, reduced dependence on remote cloud infrastructure, and applications involving IoT devices [1].

The literature also identifies several implementation challenges, including energy efficiency, maintenance and updates, scalability, and heterogeneous edge devices [1].

B. AI-Based Traffic Signal Control

AI-based traffic signal control has been investigated using different machine-learning and reinforcement-learning approaches. Rasheed *et al.* provide a review of deep reinforcement learning approaches for traffic signal control [3].

Machine-vision approaches have also been investigated for adaptive traffic-light control. Such approaches use information obtained from road scenes to support traffic signal decisions [4].

C. Edge-Based Traffic Processing

The reviewed Edge AI literature identifies transportation as an application area and discusses edge-based traffic-light data processing [1]. Wu *et al.* presented an edge-computing-based data detection scheme for traffic lights at intersections [2].

These studies provide motivation for investigating local processing as part of a traffic signal control architecture.

V. PROPOSED SYSTEM

The proposed architecture consists of five main components:

- 1) Roadside Cameras/Sensors
- 2) Edge AI Processing Unit
- 3) Traffic-State Estimation Module
- 4) Signal Decision Module
- 5) Traffic Signal Controller

An optional cloud layer can be used for long-term storage, system monitoring, model management, and large-scale analysis. The immediate traffic-control process is conceptually performed near the road.

VI. SYSTEM FLOW

The proposed system operates through the following sequence:

Traffic Data → Local Processing → Traffic-State Estimation → Signal Decision → Signal Update

The detailed flowchart is shown in Fig. 2.

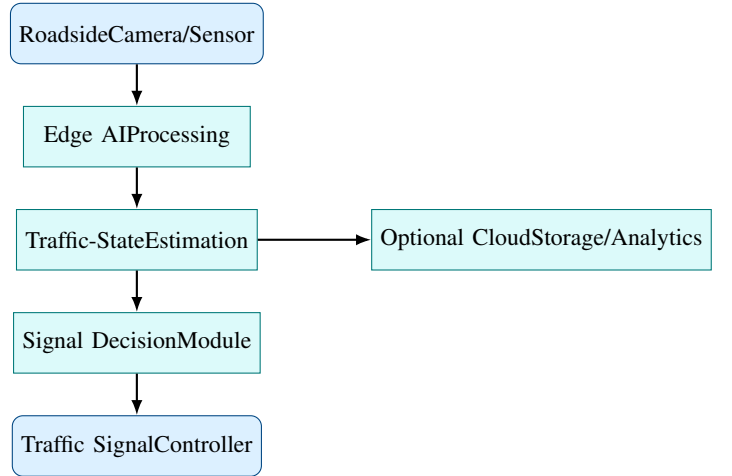


Fig. 1. Conceptual architecture of the proposed Edge AI-based traffic signal system.

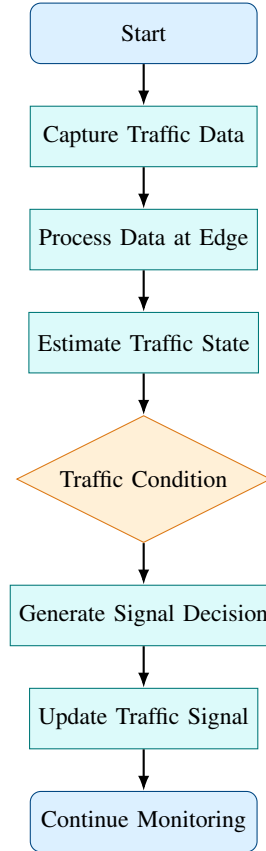


Fig. 2. Flow of the proposed Edge AI-based traffic signal system.

VII. TRAFFIC-STATE REPRESENTATION

For a conceptual model, the traffic condition at an intersection can be represented using variables such as vehicle count, estimated queue length, lane occupancy, and waiting time.

Signal Decision → Signal Update

$$X = \{x_1, x_2, \dots, x_n\} \quad (1)$$

where each x_i represents an observed traffic feature. The traffic-state estimation can be represented generally as

$$S = f(X) \quad (2)$$

where S is the estimated traffic state and f represents the traffic-analysis model.

A signal-control action can then be represented as

$$A = g(S) \quad (3)$$

where A represents the selected signal action and g represents the decision mechanism.

These equations describe the proposed processing model and are not experimental performance equations.

VIII. EDGE AI PROCESSING

The edge unit receives traffic information from roadside cameras or sensors. Instead of continuously transmitting raw data to a remote server, the edge unit can perform local processing.

A conceptual processing pipeline is:

$$X \rightarrow f(X) \rightarrow S \rightarrow g(S) \rightarrow A$$

where:

- X = collected traffic data,
- $f(X)$ = local traffic-data processing,
- S = estimated traffic state,
- $g(S)$ = signal decision process,
- A = traffic signal action.

The specific AI model can be selected according to the available hardware, dataset, computational resources, and required accuracy.

IX. SYSTEM MODULES

A. Traffic Data Collection

Roadside cameras or sensors provide information about the traffic environment. The collected information may include vehicle presence, traffic density, and lane-level observations.

B. Edge Processing

The edge computing unit processes the incoming information close to the traffic intersection. This approach is consistent with the Edge AI principle of processing data near its source [1].

C. Traffic-State Estimation

The processed information is converted into a representation of the current traffic condition.

D. Signal Decision

The signal decision module determines a suitable control action based on the estimated traffic state.

E. Signal Controller

The selected control action is communicated to the traffic signal controller.

TABLE I
CONCEPTUAL COMPARISON OF TRAFFIC SIGNAL APPROACHES

Feature	Fixed-Time Control	Edge AI Control
Decision basis	Predefined timing	Observed traffic data
Local processing	Limited	Proposed
AI-based analysis	Not required	Possible
Cloud dependency	Not required	Optional
Real-time adaptation	Limited by timing plan	Design objective
Scalability	Depends on deployment	Requires resource management

F. Cloud Monitoring

The cloud layer is optional in the proposed architecture. It can support storage, monitoring, model management, and longer-term traffic analysis.

X. CONCEPTUAL COMPARISON

The proposed architecture can be compared with conventional timed control and centralized processing at a conceptual level.

This table is a qualitative comparison. It does not claim measured performance superiority.

XI. SECURITY AND PRIVACY CONSIDERATIONS

Edge AI can reduce the need to transmit raw data to centralized servers, but local processing does not automatically guarantee security or privacy.

The proposed system should therefore consider:

- authentication of edge devices,
- secure communication,
- protection of stored traffic data,
- secure software and model updates,
- access control,
- monitoring for abnormal system behavior.

Security is particularly important because distributed edge devices may be deployed at multiple physical locations. The Edge AI literature identifies security, maintenance, and scalability as important implementation challenges [1].

XII. IMPLEMENTATION CONSIDERATIONS

An actual implementation would require suitable hardware, software, traffic datasets, and a traffic-signal interface.

Possible components include:

- roadside camera or traffic sensor,
- edge computing device,
- lightweight machine-learning model,
- traffic-state estimation software,
- traffic signal controller,
- optional cloud platform.

The selection of an AI model should consider model size, computational requirements, available memory, energy

consumption, and required accuracy. The reviewed Edge AI literature emphasizes the importance of resource management and ML model sizing for edge environments [1].

XIII. LIMITATIONS

This paper presents a proposed architecture rather than a completed experimental system. Therefore, the following limitations apply:

- 1) No real-world traffic dataset has been evaluated in this sample paper.
- 2) No measured latency values are reported.
- 3) No measured accuracy values are reported.
- 4) No cost-performance comparison is claimed.
- 5) The proposed signal-control policy requires implementation and testing.
- 6) Hardware requirements may vary according to the selected AI model and traffic environment.

These limitations are intentionally stated to avoid presenting assumed or fabricated experimental results.

XIV. FUTURE WORK

Future research can implement the proposed architecture using a real traffic dataset and an edge computing device.

Possible future work includes:

- collecting real intersection traffic data,
- training and evaluating suitable ML models,
- comparing lightweight and larger models,
- measuring processing latency,
- measuring energy consumption,
- evaluating traffic-state estimation accuracy,
- testing different signal-control strategies,
- evaluating the system under different traffic conditions,
- investigating emergency-vehicle coordination,
- studying secure model and device updates.

The Edge AI literature also identifies future research needs related to resource management, scalability, security, and real-time intelligent transportation applications [1].

XV. CONCLUSION

This paper presented a proposed Edge AI-based architecture for adaptive traffic signal control. The architecture places traffic data processing close to the intersection and uses an edge AI unit to estimate traffic conditions and generate signal-control decisions.

The approach is motivated by research showing that Edge AI can support real-time processing and is applicable to transportation and traffic-light systems. However, the proposed architecture should not be considered experimentally validated. Actual performance must be established through implementation, controlled experiments, and real-world evaluation.

Future work should therefore focus on implementing the system, selecting appropriate lightweight AI models, measuring latency and accuracy, and evaluating resource, security, and scalability requirements.

REFERENCES

- [1] S. S. Gill, M. Golec, J. Hu, M. Xu, J. Du, H. Wu, G. K. Walia, S. S. Murugesan, B. Ali, M. Kumar, K. Ye, P. Verma, S. Kumar, F. Cuadrado, and S. Uhlig, "Edge ai: A taxonomy, systematic review and future directions," *arXiv preprint arXiv:2407.04053*, 2024.
- [2] L. Wu, R. Zhang, R. Zhou, and D. Wu, "An edge computing based data detection scheme for traffic light at intersections," *Computer Communications*, vol. 176, pp. 91–98, 2021.
- [3] F. Rasheed, K. L. A. Yau, R. M. Noor, C. Wu, and Y.-C. Low, "Deep reinforcement learning for traffic signal control: A review," *IEEE Access*, vol. 8, pp. 208 016–208 044, 2020.
- [4] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, "Optimization of adaptive traffic light control modes based on machine vision," *Transportation Research Procedia*, vol. 57, pp. 241–249, 2021.